Optimal sizing of grid connected microgrid in Morocco using Homer Pro

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Abstract— **To support the international efforts climate change mitigation, Morocco is committed to rise renewable energy, to achieve energy saving and to mitigate Greenhouse Gas (GHG) emissions. Morocco has engaged to supply rural areas with access to electricity dependent on the use of available Renewable Energy Sources (RES) to diminish GHG emissions. Appropriate combination of generation units and storage devices will lead to optimal system design in terms of system technical feasibility and cost. In this paper, the objective is to optimize a MG sizing for rural community buildings in Grid Connected (GC) operation mode, based on technical and economic analysis. The MG under study includes Photovoltaic (PV), Wind Turbine (WT) and Diesel Generator (DG) as generation resources and Battery Bank (BB) as the storage devices. The load data are corresponded to a rural community buildings, and the solar radiation and wind speed data are corresponded to the city of Tahla, Morocco. Simulation results outline the effectiveness of the proposed approach for optimal conception of a GC-MG.**

Keywords—Microgrid; Sizing; Renewable Energy; Homer Pro;

I. INTRODUCTION

A. Motivations for Green power development in Morocco

Morocco is assumed to be within the countries, which are threatened by climatic change under the prediction of global warming and climatic change. Morocco's GHG emissions from fuel combustion are expected to increase quickly on the grounds of the growth of both the energy and residential sectors [1]. To support the international efforts climate change mitigation, Morocco had approved to Kyoto protocol, Paris Agreement, and had organized the UNFCCC COP 22 in Marrakech [2]. In the Middle East and North Africa (MENA), Morocco's Renewable Energy (RE) targets are the most ambitious with its national adoption of green power plan called national energy strategy. The lack of conventional hydrocarbons encourages the reliable alternative energy sources development in Morocco, such as wind power, solar power and hydro turbines [3]. Morocco is committed to rise the capacity of RE to 42% by 2020 and to 52% by 2030 of total installed capacity, to achieve energy saving of 12% by 2020 and 15% by 2030 and to attenuate GHG emissions in the sector of transport by 35% [4].

Since the 1990s, Morocco has engaged to supply remote rural areas with access to electricity dependent on the use of available local resources. The Rural Electrification Programme (PERG) has been founded to cater rural community needs based on the long-term economic profit of village- based electricity access. In areas where the connection to the grid is evaluated uneconomical, accordingly to local environment conditions, the PERG uses alternative energy such as solar power, WT, small hydro turbines, DGs and hybrid systems. [5]. This paper proposes an optimal sizing of MG with hybrid system to supply a rural village in compliance with the PERG objectives.

B. Literature Survey

For the integration of hybrid system in MG, several studies was done to achieve economic purposes by developing a new methods for Energy Management System (EMS) and sizing of generation units and storage devices for a given MG. For researches who are interested in EMS, reference [6] given a review on the MG-EMS which has discussed in recent frameworks, focused on nature of MG generation units and storage devices, the integration of electric vehicles and combined heat and power systems, and the objective functions and constraints of MG-EMS with the applied optimization algorithms. This paper discusses the issue of optimal sizing of MG components and below some of the previous research paper that have proposed several methods to solve this problem. Reference [7] analyzed a bi-level optimization method combined sizing and energy management of stand-alone MG, which includes PV panels, Battery Storage System (BSS) and Hydrogen Storage System (HSS). The HSS under study comprise of electrolyzer, hydrogen tanks and a Fuel Cell (FC) units. The author has considered Operation and Maintenance (O&M) cost and the annual capital cost as economical criterions to obtain optimal sizing of MG units. For the operation cost of BSS, the author has taken in consideration the Investment Cost (IC) and the degradation of the BSS. Reference [8] proposed energy management and control system for GC-MG, the required BSS is sized to ensure energy supply during 3 h load shedding period, for a facility of a peak load, with BSS depth of discharge of 80% to extend the life of BSS with efficiency of 95%. The PV system is oversized to take in consideration loss power and winter season. The system is tested under standalone operation mode, during 24 h time horizon to ensure optimal use of PV and BSS for all periods over the studied horizon. Reference [9] analyzed an optimal energy storage sizing for an islanded PV/battery MG supplying ten residential customer's demand. The system is solved using mixed-integer linear programming. The results demonstrate that ESS can improve electric power reliability compared to the system without a centralized ESS. Moreover, a smaller ESS size can be applied for centralized ESS strategy with a large amount of load

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supplied and solar utilization. Reference [10] discussed both the sizing and control of a standalone MG including wind energy and energy storage system. The used energy storage sizing method has taken into account reliability requirement of MG. The capital cost of ESS was the main factor of sizing mythology as well as its annualized cost (including capital and maintenance costs) and compensation costs of curtailed load and wind power. Reference [11] presented the sizing of standalone MG with wind power, solar power, FC, and BB. The author has simulated the behavior of various configurations of RES based on technical and economic analysis including system initial IC, O&M cost and cost of energy. The author has concluded that the combination of WT with PV and BB are less costly compared to other configuration with a FC. Reference [12] presented multi-objective line-up competition algorithm to optimize a standalone MG configuration. The author justified that the combination of PV/WT/DG system is optimal than the PV/WT. The annual maintenance cost and the annual capital cost are taken as the economic factors to obtain the optimal MG configuration. Reference [13] proposed to optimize the sizing of an isolated MG including PV/WT/ DG /BSS. A discrete version of harmony search is used as sizing optimization method. The proposed combination is compared to a DG power system in term of annual maintenance cost of PV/WT and DG. However, the maintenance costs of battery is neglected. Reference [14] used a genetic algorithm to optimize GC-MG including PV and BSS, the analysis factors were the economic and environmental goals represented in term of the system's self sufficiency ratio and net present value. The net present value includes the system IC, O&M cost, Replacement Cost (RC) and system revenues within the system lifetime. Reference [15] presented the design of GC-PV system in Makkah, Saudi Arabia. The optimal PV array and inverter sizes were investigated based on several criterions such as fraction of renewable electricity, excess electricity, Net Present Cost (NPC) and Carbon Dioxide (CO2) emissions.

The previous papers have implemented a complex methods to design the optimal MG configuration, and some of these have neglected a considerable cost factor into sizing analysis. This paper uses Homer Pro the last version of Homer software that is known as a powerful optimization tool. Homer Pro is available for researchers to download free of charge. It can perform optimization design over a wide range of energy generators, energy storage devices, converters and loads. Moreover, Homer can simulate a data of a year in time-intervals of up to 1 min [16].

II. THE COMPONENTS MODELS OF MICROGRID

A. Wind Turbine Model

The output power of WT generator depends on wind speed condition and can be measured according to (1). The cut-in, cut-out and nominal speed of WT are considered as crucial parameters for the determination of WT output power.

$$P_{WT}(t)=\begin{cases} a*V^3(t)-b*P_r & V_{ci}\prec V\prec V_r \\ P_r & V_r\prec V\prec V_{co} \\ 0 & V\prec V_{ci}\ \&\ V_{co}\prec V \end{cases} \quad (1)$$

The variables a and b are calculated by (2) and (3) [17]:

$$a=P_r/(V_r^3-V_{ci}^3) \quad (2)$$

$$b=V_{ci}^3/(V_r^3-V_{ci}^3) \quad (3)$$

Where P_r is the rated output power of WT, V_r is the nominal wind speed, and V_{co} and V_{ci} are the cut-out speed and cut-in speed, respectively.

B. Photovoltaic Module Model

The output power of PV module can be described by (4):

$$P_{PV}(t)=\eta_m*\eta_{conv}*A*G(t)*(1-\beta_{th}(T_c(t)-T_r)) \quad (4)$$

Where:

η_m is the efficiency of PV module, η_{conv} is the convertor's electrical efficiency, A are the total surface of PV panels (m^2), G is the total solar radiation (W/m^2), β_{th} is thermal factor, T_r is the reference temperature of PV module. And the cell temperature T_c is defined by (5):

$$T_c(t)=T_a(t)+G(t)*(NOCT-20)/800 \quad (5)$$

Where: T_a is the ambient temperature, and NOCT is the nominal cell temperature.

The PV energy is defined by (6):

$$E_{PV}(t)=P_{PV}(t)*\Delta t \quad (6)$$

C. Battery Model

The energy of battery can be modeled using (7) [17]:

$$E_B(t)=V_B*C_B*(SOC(t-1)-SOC(t)) \quad (7)$$

Where, V_B is the battery's nominal voltage, C_B is the battery's nominal capacity, and SOC is the battery's state of charge.

Lithium-Ion (LI) battery is used in this paper due to its advantages in terms of efficiency, power density and energy, temperature operating, and lifespan [18].

The battery's *SOC* of LI battery is estimated by (8) according to ampere-hour counting approach [19]:

$$SOC(t)=SOC(0)-\frac{T}{C}\int_0^t(\eta*I(t)-\delta)dt \quad (8)$$

Where SOC(0) is the battery's initial SOC, I(t) is the current, T is the sampling period, C is the battery's nominal capacity, η is the coulombic efficiency, and δ is self discharge.

III. HOMER SOFTWARE DESCRIPTION

Homer Pro is MG optimization software based on economic and technical assessment, originally designed by National Renewable Energy Laboratory (NREL), and improved by Homer energy. Homer Pro is developed to design both GC and isolated hybrid MG using renewables and conventional sources to serve electric and thermal loads [20]. Homer Pro needs as inputs the resource data like load profile to be supplied, average daily environment condition (solar irradiance or/and wind speed), generating and storage units to be implemented and the corresponding costs [21]. Homer Pro software models the physical behavior of hybrid MG units and its life cycle cost (including capital cost, O&M costs, RCs and fuel cost). Homer Pro allows comparing different designs of generating and storage units based on their economic and technical criteria. The software then examines an hourly power balance calculation for each generation and storage units configuration for a year [21]. The objective is to determine the best configuration that has a lowest total NPC that can be described by (9) and (10) [17]. Then, when all the possible configurations are simulated, the feasible solutions are classified depending on the total NPC.

$$C_{NPC} = \frac{C_{total\_annual}}{CRF(i,R)} \tag{9}$$

$$CRF(i,R) = \frac{i(1+i)^R}{(1+i)^R - 1} \tag{10}$$

Where C_{total_annual} is the total annual cost, i is the real interest rate, and R is the project lifetime.

IV. CASE STUDY DESCRIPTION AND HOMER PRO INPUTS

To perform economic analysis and sizing of MG, a rural MG located around Tahla city, Morocco is considered. Fig. 1 shows the hourly profile of electrical Load Demand (LD) in an ordinary day, and Fig. 2 shows the average of daily LD in various months of a year. The load has a scaled annual average demand of 165.59 kW/day with scaled peak of 29.25 kW taking in consideration as random variability of 10% day to day for 20% time step.

Considered rural community building is located at latitude of 34.05(north), longitude of 4.42(west), and altitude of 598 meters. The nature of RES affects directly the output and economy of the system. Clearness index shown in Fig. 3 can be measured by (11) [22]:

$$Kt = \frac{H_h}{H_0} \tag{11}$$

Where H_h is the horizontal global irradiance and H_0 is the irradiance available outside the atmosphere.

In this paper, RES will be implemented with backup DG and BB to serve the rural community building in GC operation mode. The new MG architecture is designed to reduce grid dependency and increase economic viability of the system. The excess of generated power will be stored in BB.

The place has a monthly average daily solar radiation of 4,972 kWh/m²/day with minimum of 2.470 kWh/ m²/day in December and maximum of 7.390 kWh/m²/day in July. Fig. 3 shows the average of daily solar radiation and clearness index of Tahla in various months of a year. The average hourly wind speed is measured at 10 meters above the ground and the data of the year 2017 is provided by NASA's MERRA-2 Modern-Era Retrospective Analysis [23]. The average hourly wind speed at the hub height of WT can be defined by (12). The place has an average daily wind speed of 3.09 m/s with maximum of 8.40 m/s in January. Fig. 4 shows monthly average wind speed obtained.

$$v_2 = v_1 * \frac{\ln(\frac{h_2}{z_0})}{\ln(\frac{h_1}{z_0})} \tag{12}$$

Where the reference wind speed v_1 is measured at the reference height h_1 in this paper h_1=10m. v_2 is the wind speed at height h_2. z_0 is the roughness length.

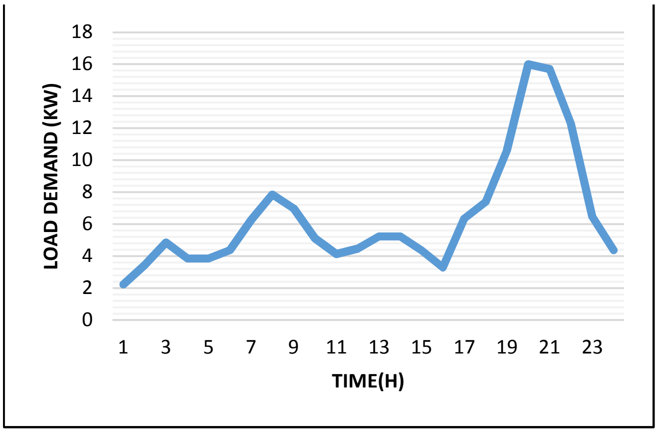

Fig. 1. The daily load demand

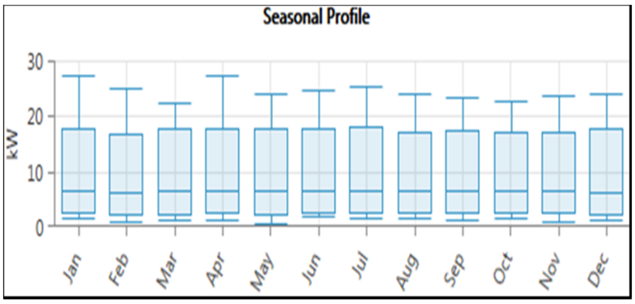

Fig. 2. The average of daily load demand in various months of a year

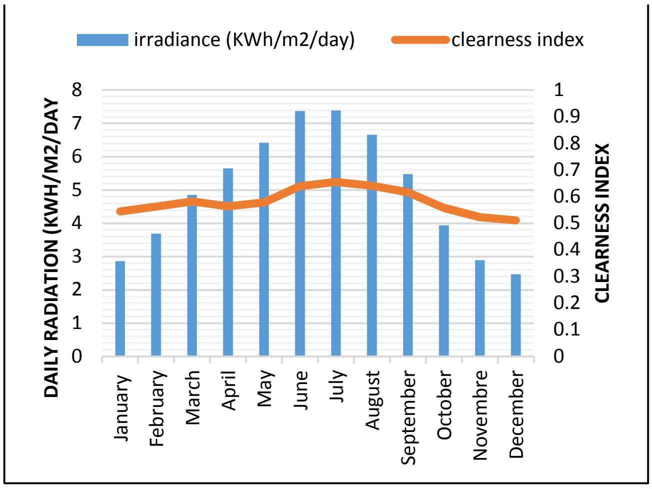

Fig. 3. The average of daily solar radiation and clearness index of Tahla in various months of a year

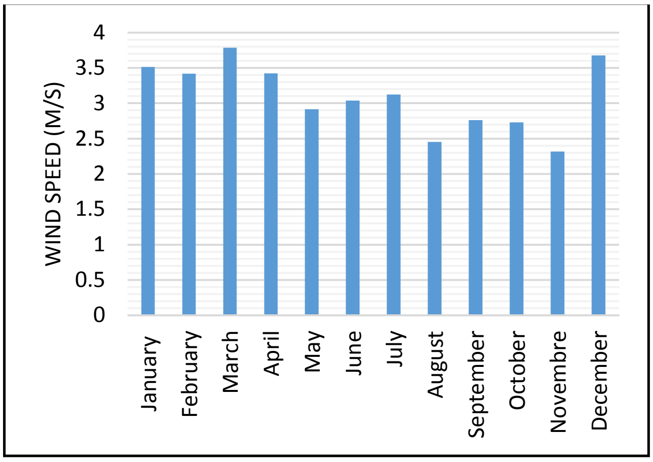

Fig. 4. The average of daily wind speed of Tahla in various months of a year

V. MICROGRID MODELING AND SIMULATION RESULTS

A. Optimization Model

Each month of the year, the simulation considers into study the solar radiation, the clearness index, the orientation and location on Earth's surface. For PV model, the Initial Capital Cost (ICC) is taken as 640$/kW, RC as 640$/kW, and O & M cost as 10$/kW per year as suggested by [24]. Lifetime of the PV is 25 years with 80% derating factor. For WT model, with rated power of 3kW, the ICC is taken as 1690$ [25], RC as 1690$, and O & M cost as 17$ per year. Lifetime of the WT is 20 years with hub height of 17m.

Various combinations of distributed resources are simulated to reach the optimal MG design. The performance evaluation is based on the minimization of the objective function subject to the system constraints. The objective function is the minimization of the total NPC that defined as the HOMER's main economic output [21]. The total NPC is the present value of the sum of all costs over the system lifetime, minus its earned revenue. The costs include the cost of buying energy from the grid, capital costs, RCs, O&M costs, fuel costs as well as the emissions penalties. The revenues are salvage value and the revenue of energy sold to the grid [21]. Homer used the value of the total NPC to rank all MG configurations in the optimization results. This paper implements PV, WT, DG, battery, and converters as the components of MG in GC operation mode. According to the minimum NPC, Homer software decides the finest feasible MG design that can effectively serve the loads. Fig.5 describes overall MG architecture obtained by interconnecting all the distributed units with respect to the system modelling specifications expressed in TABLE I. The project lifetime is fixed at 25 years. For the major constraints, the operating reserve as percentage of loads is set at 7% for Load in current time step and at 5% for Annual peak load. The operating reserve as percentage of renewable output is set at 20%. Minimum renewable fraction is set at 35%. Maximum annual capacity shortage is fixed at 20%. To take in consideration the environment pollution and emission reduction, emissions penalties are introduced as 20 $/ton for Carbon dioxide, 20 $/ton for Sulfur dioxide, and 20 $/ton for Nitrogen oxides. The project takes in consideration the effects of the environment condition uncertainties as sensitivity parameters. For the study, wind speed and solar radiation are varied 90%-110% of scaled annual average value [26].

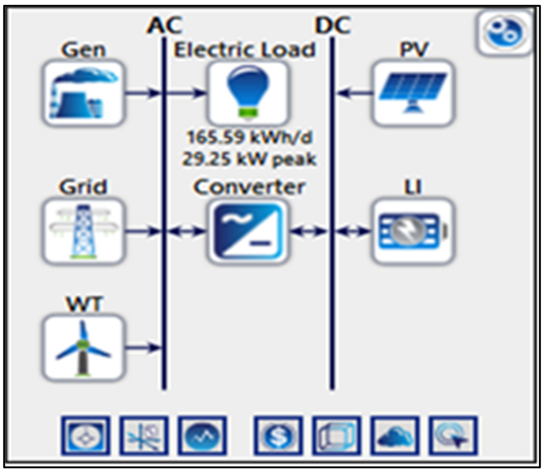

Fig. 5. MG architecture

TABLE I. MICROGRID SYSTEM MODELING SPECIFICATIONS

Component	Capital cost	RC	O&M cost	Lifetime	Efficiency	Other parameters
WT generator	1690 $	1690 $	17$/year	20 year	----	Hub Height: 17 (m)
PV Array	640 $/kW	640 $/kW	10 $/kW	25 year	----	Derating Factor: 80%
Converter	375 $/kW	375 $/kW	10 $/kW	25 year	97%	----
Battery Device	550 $/kW	550 ($/kW)	10 $/kW	15 year	90 %	----
DG	500 $/kW	500 $/kW	0.03 $/kW/hour	15000 hour	-----	----

B. *Simulation Results*

In the optimization process with HOMER Pro, several configurations of the MG are simulated and total NPC is calculated. The optimal MG configuration is the best combination with the lowest NPC. TABLE II represents the simulation results using HOMER Pro for the study rural MG in Tahla area. It can be seen that the implementation of WT and BB with PV/DG/Converter reduces the Total NPC and the cost of O&M and increases the renewable fraction in the MG. The combination of PV/WT/DG/Battery/Converter is more cost effective than PV/DG/Converter and PV/WT/DG/Converter. In the simulation result, the mean generated power by RES is equal to 46736 (kWh/yr) that represents 53.4% of total energy production in MG, is the most share of energy in MG. It can be noted that generation of RES is high in spring and summer seasons depending on Tahla environment conditions. Fig. 6 represents the NPC summary resulted from the Homer Pro optimization, the highest ICC is about 20892$ that related to purchasing cost of PV. However, the most RC within 25 years of lifetime is 1867$, which related to BB. The highest operation cost is 65209$ representing the grid operation and the only cost of fuel is about 2334$ and related to DG.

Fig. 2 shows the seasonal profile of MG load profile, it can be observed that MG required more energy in summer and winter seasons. The implementation of RES can notably reduce the grid dependency. Homer Pro optimization result allow MG to reduce net energy purchased from grid that can be defined as energy purchased from grid minus energy sold to grid. As can be seen in Fig. 7 and Fig. 8, the net energy purchased is clearly reduced in summer period, when the RE produced is more important. The daily LD is lower at the period when production of RES is higher, therefore, the excess energy can be stored in BB and the surplus is sold to the grid. Also in the hours with generation deficiency, the BB inject power to the MG for demand satisfaction. When the battery can't satisfy the LD, the MG purchases power from the grid. For example, in June the MG purchases 3063 kWh from grid when the production of RES is not enough to supply the LD and sells 2348 kW to grid when production of RES exceeds load consumption. Therefore, the net purchased energy from the grid is 715 kWh, which means to pay 354.56 $.

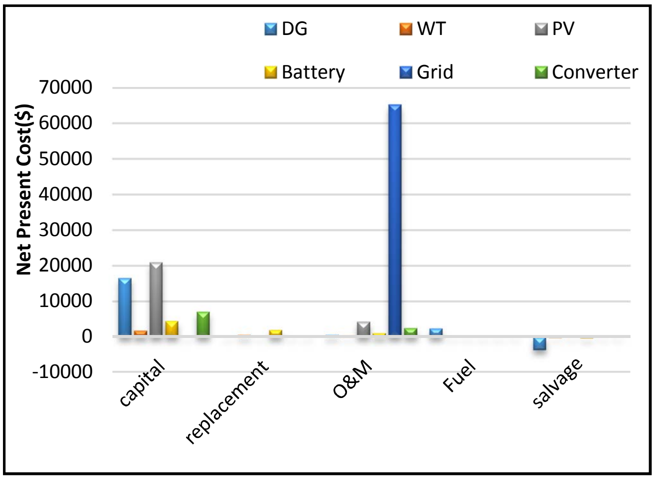

Fig. 6. NPC summary for different component of MG

TABLE II. SIMULATION RESULTS

	PV (kW)	WT (kW)	DG (kW)	Battery	Converter (kW)	Total NPC ($)	Total O&M Cost ($/yr)	Operating Cost ($/yr)	COE ($)	Renewable Fraction	Fuel (L/yr)
1	32.6	3	33	8	18.7	131312	42.6	6.252	0.121	51.4	181
2	32.6	--	33	--	18.5	139470	345	7.363	0.129	50.9	1455
3	32.2	3	33		18.3	141134	345	7.387	0.131	50.9	1455

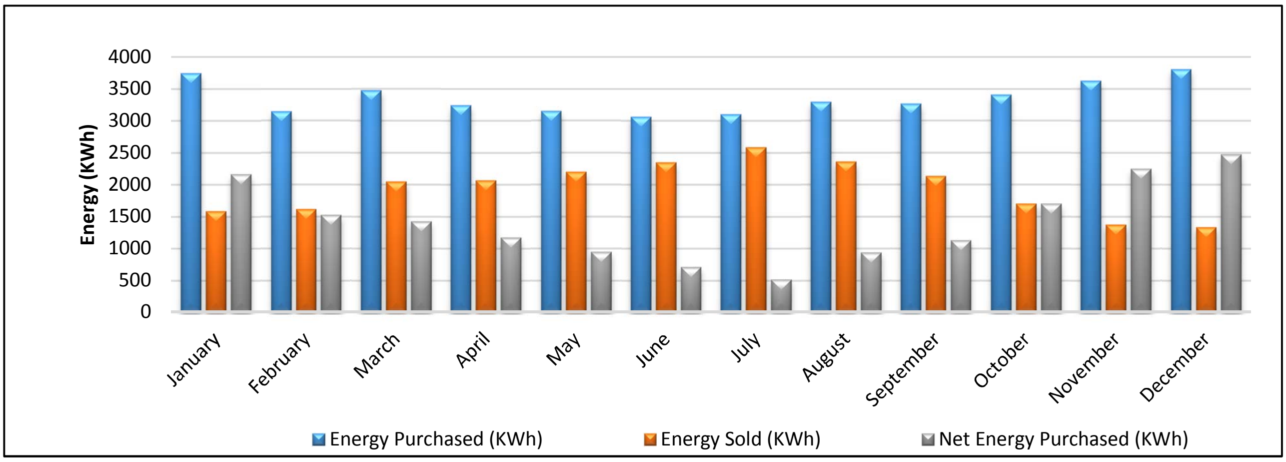

Fig. 7. The energy exchanged between the grid and the MG

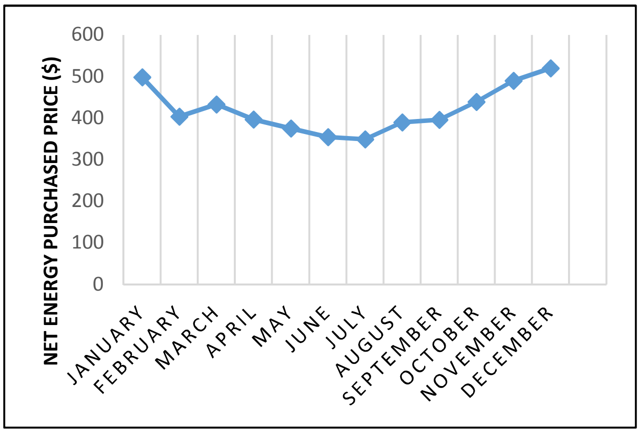

Fig. 8. The cost of net energy purchased from the grid.

VI. Conclusion

In this paper, optimal design of a grid connected microgrid is described. The microgrid is consisted of wind turbine, photovoltaic, battery and diesel generator. The considered microgrid is a rural building community and the solar radiation and wind speed data are corresponded to the city of Tahla, Morocco. The optimal sizing are achieved by using HOMER PRO. The simulation results outline that the highest initial capital cost is about 20892$, which presented the purchasing cost of photovoltaic system. However, the most replacement cost within 25 years of lifetime is 1867$, which related to battery bank. The highest operation cost is 65209$ representing the grid operation and the only cost of fuel is about 2334$ and related to DG. The load demand is mainly fed by renewable sources with 46736 (kWh/yr), which represents 53.4% of total energy production in microgrid.

References

[1] Kousksou, T., Allouhi, A., Belattar, M., Jamil, A., El Rhafiki, T., Arid, A., & Zeraouli, Y. (2015). Renewable energy potential and national policy directions for sustainable development in Morocco. *Renewable and Sustainable Energy Reviews*, *47*, 46-57.

[2] Daadaoui, L., & Saoud, S. (2017). Understanding the determinants of ecological Behavior of Young Moroccans. to appear.

[3] Hochberg, M. (2018). Renewable Energy Growth in Morocco: An Example for the Region. 26, MEl Policy Focus 2016.

[4] AZEROUAL, M., EL MAKRINI, A., EL MOUSSAOUI, H., & EL MARKHI, H. (2018). Renewable Energy Potential and Available Capacity for Wind and Solar Power in Morocco Towards 2030. Journal of Engineering Science & Technology Review, 11(1).

[5] El-Katiri, L. (2016). The Road to Marrakech: Key Issues for COP22/La Voie vers Marrakech: Quelles Priorités pour la COP22?.

[6] Boqtob, O., El Moussaoui, H., El Markhi, H., & Lamhamdi, T. (2019). Microgrid energy management system: a state-of the-art review. Journal of Electrical Systems, 15(1), 53-67.

[7] Li, B., Roche, R., & Miraoui, A. (2017). Microgrid sizing with combined evolutionary algorithm and MILP unit commitment. *Applied energy*, *188*, 547-562.

[8] Syed, I. M., & Raahemifar, K. (2016). Predictive energy management and control system for PV system connected to power electric grid with periodic load shedding. *Solar Energy*, *136*, 278-287.

[9] Habib, A. H., Disfani, V. R., Kleissl, J., & de Callafon, R. A. (2017, May). Optimal energy storage sizing and residential load scheduling to improve reliability in islanded operation of distribution grids. In *American Control Conference (ACC), 2017*(pp. 3974-3979). IEEE.

[10] Luo, Y., Shi, L., & Tu, G. (2014). Optimal sizing and control strategy of isolated grid with wind power and energy storage system. Energy Conversion and Management, 80, 407-415.

[11] Hosseinalizadeh, R., Shakouri, H., Amalnick, M. S., & Taghipour, P. (2016). Economic sizing of a hybrid (PV–WT–FC) renewable energy system (HRES) for stand-alone usages by an optimization-simulation model: case study of Iran. Renewable and Sustainable Energy Reviews, 54, 139-150.

[12] Shi, B., Wu, W., & Yan, L. (2017). Size optimization of stand-alone PV/wind/diesel hybrid power generation systems. *Journal of the Taiwan Institute of Chemical Engineers*, *73*, 93-101.

[13] Maleki, A., & Askarzadeh, A. (2014). Optimal sizing of a PV/wind/diesel system with battery storage for electrification to an off-grid remote region: A case study of Rafsanjan, Iran. *Sustainable Energy Technologies and Assessments*, *7*, 147-153.

[14] Zhang, Y., Lundblad, A., Campana, P. E., Benavente, F., & Yan, J. (2017). Battery sizing and rule-based operation of grid-connected photovoltaic-battery system: A case study in Sweden. *Energy conversion and management*, *133*, 249-263.

[15] Ramli, M. A., Hiendro, A., Sedraoui, K., & Twaha, S. (2015). Optimal sizing of grid-connected photovoltaic energy system in Saudi Arabia. Renewable Energy, 75, 489-495.

[16] Fathima, A. H., & Palanisamy, K. (2015). Optimization in microgrids with hybrid energy systems–A review. *Renewable and Sustainable Energy Reviews*, *45*, 431-446.

[17] Shahinzadeh, H., Moazzami, M., Fathi, S. H., & Gharehpetian, G. B. (2016, December). Optimal sizing and energy management of a grid-connected microgrid using HOMER software. In Smart Grids Conference (SGC), 2016 (pp. 1-6). IEEE.

[18] Hannan, M.A.; Lipu, M.S.H.; Hussain, A.; Mohamed, A. A review of lithium-ion battery state of charge estimation and management system in electric vehicle applications: Challenges and recommendations.nRenew. Sustain. Energy Rev. 2017, 78, 834–854.

[19] Zhang, R., Xia, B., Li, B., Cao, L., Lai, Y., Zheng, W., ... & Wang, W. (2018). State of the art of lithium-ion battery soc estimation for electrical vehicles. Energies, 11(7), 1820..

[20] Rajbongshi, R., Borgohain, D., & Mahapatra, S. (2017). Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER. Energy, 126, 461-474.

[21] Homer Microgrid NEWS and Insight. Available online: http://microgridnews.com/homer-optimizer-afaster-path-to-finding-least-cost-microgrid-options/ (accessed on 23 August 2017).

[22] Ineichen, P., Guisan, O., & Razafindraibe, A. (1984). Mesures d'ensoleillement à Genève: publication No 9: Indice de clarté.

[23] https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/

[24] Al Garni, H. Z., Awasthi, A., & Ramli, M. A. (2018). Optimal design and analysis of grid-connected photovoltaic under different tracking systems using HOMER. *Energy Conversion and Management*, *155*, 42-57.

[25] Alibaba. 3kw portable small horizontal wind turbine price. https://www.alibaba.com/product-detail/Chinese-3kw-portable-small-horizontal-wind_60482340445.html?spm=a2700.7735675.normalList.1.EmNZqw&s=p

[26] Kumar, Y. P., & Bhimasingu, R. (2014, December). Optimal sizing of microgrid for an urban community building in south India using HOMER. In *Power Electronics, Drives and Energy Systems (PEDES), 2014 IEEE International Conference on* (pp. 1-6). IEEE.